

# Machine Learning Pipeline Workflow

## STANDARD OPERATING PROCEDURE & EXECUTION FRAMEWORK

### Part I: Data Understanding & Initial Assessment

#### 1. Data Sampling

- Count numerical and categorical features
- Check and inspect missing data
- Drop unnecessary features

#### 2. Data Information

- Execute `info()` to inspect metadata and data types

#### 3. Statistical Summary ( `describe()` )

- Analyze percentile values and compare with data distribution graphs
- Analyze percentile-wise changes in standard deviation (std) and mean

#### 4. EDA Plotting

- Analyze graphs for all features with respect to the target variable
- Analyze feature-to-feature graphs (if needed)

### Part II: Data Preprocessing & Modeling

#### 1. Outlier Detection

- Identify and handle anomalous data points

#### 2. Column Transformation

- **Scaling:** Normalize/Standardize continuous numerical variables
- **Encoding:**
  - *Nominal Encoder:* Used when there is no natural order
  - *Ordinal Encoder:* Follows a natural order of categories
  - *Label Encoder:* Converts categorical targets/labels into integers
- **Imputation:** Fill missing values using standard techniques

### 3. Dimensionality Reduction (PCA)

- Apply Principal Component Analysis (PCA) if needed when the number of features is very large

### 4. Model Selection

- Select candidate machine learning algorithms for evaluation

### 5. Model Training

- Train selected algorithms on processed dataset

### 6. Model Evaluation

- Once best performance is achieved, check for underfitting and overfitting. If issues exist, loop back through steps 1–6

### 7. Hyperparameter Tuning

- If evaluation performance repeatedly falls short, perform hyperparameter tuning

### 8. Correlation Heatmap

- Visualize feature correlations to verify relationships

### 9. Cross-Validation

- Validate model generalization across multiple data folds

### 10. Confusion Matrix

- Analyze classification prediction errors (True/False Positives & Negatives)

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### 11. Final Verdict

Signed off by: .....

Date: .....